

Delivery Package Index — Power-Tool Battery Cell Design

Case name: `exp/t3\_r1\_noforce` — power-tool battery cell (nominal capacity ≥ 2 Ah, 5C discharge retention ≥ 95 %, 4C fast charge without plating, $T_{\text{max}} \leq 60$ °C, power density ≥ 4000 W/kg)

Numbering scheme: `VBF---` (case ID `t3\_r1\_noforce` → `T3R1NOFORCE`)

Generation date: 2026-08-25

Signature block: Prepared: _____ / Reviewed: _____ / Approved: _____

Document code reference table

Document code	Meaning	Corresponding file
DS	Specification	design_spec.md
BOM	Bill of Materials	bom.xlsx
DSH	Datasheet technical parameter sheet	datasheet.docx
CALC	Calculation sheet	calc.xlsx
DVPR	Design verification report	dvpr.md
DFMEA	Failure analysis	dfmea.md
IDX	Delivery package index (this file; package cover per verify-deliverables code table)	delivery_index.md
CAD	Structure model (optional; not requested in this task → not generated, not registered)	—

File list (only actually generated files)

File name	Number	Format	Source description
design_spec.md	VBF-T3R1NOFORCE-DS-01	md	generated per the deliverable-design-spec spec; values from params_v8.json + Chen2020 base parameter set + simulation outputs
design_spec.pdf	VBF-T3R1NOFORCE-DS-01	pdf	md → pdf release version (reportlab)
report.html	VBF-T3R1NOFORCE-DS-02	html	`bda render` from log.jsonl — seven-section audit report, ancillary to the specification
bom.xlsx	VBF-T3R1NOFORCE-BOM-01	xlsx	openpyxl; component masses from parameter-set densities/volume fractions + literature defaults (annotated)
bom.pdf	VBF-T3R1NOFORCE-BOM-01	pdf	xlsx → pdf release version (reportlab)
datasheet.docx	VBF-T3R1NOFORCE-DSH-01	docx	python-docx per the deliverable-datasheet spec; values from simulation outputs
datasheet.pdf	VBF-T3R1NOFORCE-DSH-01	pdf	docx → pdf release version (reportlab)
calc.xlsx	VBF-T3R1NOFORCE-CALC-01	xlsx	openpyxl per the deliverable-calc-sheet spec (inputs → capacity/energy → energy density → N/P & mass → process)

File name	Number	Format	Source description
calc.pdf	VBf-T3R1NOFORCE-CALC-01	pdf	xlsx → pdf release version (reportlab)
dvpr.md	VBf-T3R1NOFORCE-DVPR-01	md	virtual DVP&R; per the deliverable-dvpr spec; result values from simulation outputs
dvpr.pdf	VBf-T3R1NOFORCE-DVPR-01	pdf	md → pdf release version (reportlab)
dfmea.md	VBf-T3R1NOFORCE-DFMEA-01	md	qualitative design FMEA per the deliverable-dfmea spec (simulation-signal basis)
dfmea.pdf	VBf-T3R1NOFORCE-DFMEA-01	pdf	md → pdf release version (reportlab)
delivery_index.md	VBf-T3R1NOFORCE-IDX-01	md	package index per the deliverable-package spec
delivery_index.pdf	VBf-T3R1NOFORCE-IDX-01	pdf	md → pdf release version (reportlab)

Notes: engineering manufacturability drawings (with tolerances), material specifications and line process cards are outside the pure-simulation boundary and are not part of this package (stated honestly per protocol). CAD structure model `cell\_model.stl` is an optional deliverable decided by user clarification; it was not requested in this task and is therefore not registered.